Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “friction” and “static” to explain your answer. Sentence starters are optional.

friction: a contact force opposing relative sliding or its tendency

static: not sliding relative to the contacting surface

kinetic: associated with motion; kinetic friction acts during sliding

01 Warm-up: bring back an earlier idea

A spacecraft of mass **3 kg** pushes an asteroid of mass **300 kg** with **15 N**. **Compare interaction forces** and the accelerations due to that interaction.

Word help: interaction = two objects affecting one another

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\mathbf{0 \le f_s \le \mu_s N}$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with **3 N**. Static friction is **3 N** opposite the push, provided its maximum is at least **3 N**. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a **5 N** push that is still below the maximum.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Box remains at rest. Redraw for a 5 N push.

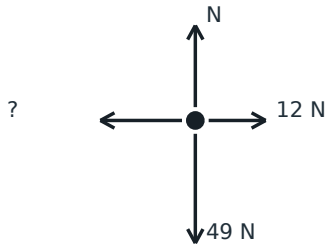
Marzano check - circle: ① ② ③

03 Actual versus maximum

A **5 kg** box is initially at rest on a level floor. Its friction coefficients are μ_s = **0.40** and μ_k = **0.30**. A horizontal push is **12 N**.
1. Use g = **9.8 N/kg**.
2. **Find the normal force**, maximum static friction, actual friction and acceleration.
3. **Label the diagram**.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Initially at rest

Marzano check - circle: ① ② ③

04 Sliding case

Increase the push on that box to **25 N**. **Find friction** and acceleration after it slides.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Estimate from data

A class tests a block with a **20 N** normal force.
1. Use the repeated measurements to **find mean peak force**, mean sliding force, and estimates of μ_s and μ_k.
2. Which estimate describes the threshold for slipping?

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Peak force (N)	Sliding force (N)
1	8	6
2	9	5
3	7	7

Marzano check - circle: ① ② ③

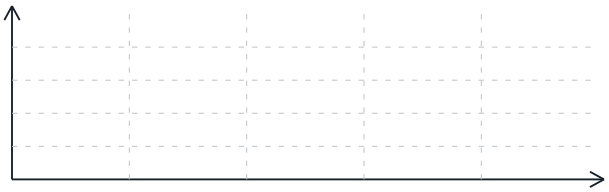
06 Review: connect an earlier idea

x_i=**-4±0.5 m**, v=**6±0.1 m/s** and exact t=**8 s**. **Give central final position**, its conservative interval and full width. **Name something you** are treating as true. **Label the diagram** with the given positions and any prediction.

Word help: final = at the end of the interval; subscript f

Sentence starter: I predict … . I used … to make this prediction because … .

Position (m)



Time (s)

Marzano check - circle: ① ② ③

07 Exit check

A **10 kg** box has μ_s = **0.50** and μ_k = **0.40**.
1. Use g = **9.8 N/kg**.
2. **Compare a 30 N** push with a **60 N** push, starting from rest each time.
3. **Find actual friction** and acceleration in both cases.

Sentence starter: Both … . They differ in … because … .

Push (N)	Still or sliding?	Friction (N)	a (m/s²)
30	____	____	____
60	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **5 kg** box has μ_s=**0.4** and μ_k=**0.3** on a level floor; use g=**10 N/kg**. A horizontal **22 N** pull begins from rest. **Find maximum static friction**, acceleration after motion starts, and the pull needed for steady sliding.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: ① ② ③